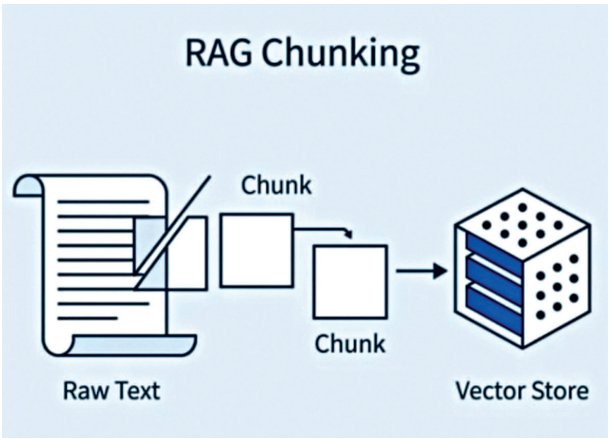
Replace `YOUR_INSTALLED_MODEL` with the exact model name shown by your runtime. Make sure the local server is running, then execute `python src/ask.py`. Ask something whose answer exists in your documents. If the model answers confidently but the retrieved passages are irrelevant, the generation step is not your first problem: retrieval is.

Chunking determines what can be found

A long document cannot always be represented well by one vector, while microscopic chunks lose the context around a rule. Chunking is therefore an editorial decision as much as a computational one.



Fixed-size chunks are easy to implement and benchmark. Sentence chunks preserve grammatical boundaries but may be too short. Paragraph chunks keep local context. Recursive chunking tries larger boundaries first, then divides oversized sections. Structure-aware chunking respects headings, tables, lists and document sections.

Start by comparing two settings: passages of roughly 150 words with thirty-word overlap, and complete paragraphs capped at the embedding model's supported input length. Ask the same twenty questions against both indexes. Which version consistently retrieves the decisive clause, including its exception? Keep that version.

Overlap helps when an answer crosses a boundary, but excessive overlap produces nearly identical search results and bloats storage. More importantly, embedding models have token limits: words, characters and model tokens are not interchangeable. Check the chosen encoder's maximum input length and examine truncation warnings or tokenizer counts, especially for code, tables and long identifiers.

Never split a refund amount from its currency, a heading from its eligibility condition, or a table row from its column labels. A clean-looking chunk can still be semantically incomplete.

Embeddings: Search by meaning

An embedding converts text into a vector, a numerical representation that can be compared with other vectors. Similar meaning often corresponds to vectors that are close under the chosen similarity measure. This helps a question about "money back after cancellation" retrieve a passage mentioning "refund following order cancellation", even when wording differs.

Sentence Transformers offers local embedding models; some support specialised query and document encodings. When a model provides those pathways, use them consistently rather than applying an arbitrary prompt prefix. Always index and query with the same model and compatible settings. Changing the embedding model usually means rebuilding the index.

Embeddings are not knowledge or proof. They can overlook a serial number, confuse related product generations or rank an older policy above a current one. They also do not automatically translate every Indian language equally well. Test Hindi, Hinglish and domain-specific terminology with real questions. An index can be technically healthy and linguistically inadequate.

